Assumptions of Physics
Technical Brief
More technical briefs at https://assumptionsofphysics.org/resources/briefs

Unitary evolution as infinitesimal projective processes

Gabriele Carcassi for the AoP collaboration

Category: Quantum mechanics - Tags: Unitary evolution, Projective measurement, Quantum Zeno effect

Abstract

We show that unitary evolution can be characterized, in a precise sense, as the limit of repeated projective measurements. The key insight is that there exist processes that can be understood either as an exponentially weighted average of the unitary trajectory or, for each incoming ensemble, as a projective measurement adapted to that ensemble. The result holds for separable Hilbert spaces, including unitary groups with unbounded generators.

1 Introduction

Unitary evolution and projective measurement are usually presented as two distinct types of quantum processes. The first preserves pure states and is deterministic and reversible. The second generally turns a pure ensemble into a statistical mixture and is neither deterministic nor reversible. Nevertheless, unitary evolution can be recovered as the limit of projective processes, much as reversible quasi-static evolution in thermodynamics can be understood as a limit of equilibration processes. The physical intuition is that, since every state is an eigenstate of its own projector, and the projector evolves in the Heisenberg picture, the evolved state is always an eigenstate of the evolved projector. Therefore, time evolution is effectively "measuring" the evolved projector at every moment in time.

Naturally, this projective measurement must depend on the incoming ensemble. For each ensemble, the projective decomposition must be chosen so that one outcome predicts the next state with probability approaching one. At the same time, the process on ensembles must remain affine, so that its output does not depend on a particular decomposition of a mixed ensemble. We call such a process state-adaptive projective, as it is effectively a projective measurement that depends on, and is adapted to, the incoming state.

The main mathematical idea is simple. Let $\mathcal{L}$ be the generator of the unitary evolution on density operators. The finite process of duration τ is the resolvent $(I - \tau\mathcal{L})^{-1}$. This map is an exponentially weighted average of the unitary trajectory. More surprisingly, for each incoming ensemble it is also exactly a projective-measurement process. Repeating it over shorter and shorter intervals recovers the unitary group.

2 Mathematical prerequisites

We define a state-adaptive projective process as a black-box process that can be understood as performing a projective measurement on each incoming ensemble. It is state-adaptive in the sense that the projective measurement depends on the incoming state.

Definition 1 (State-adaptive projective process)**.** *A state-adaptive projective process is a process that acts as a projective measurement on each ensemble. That is, it is an affine map $\Psi : \mathcal{D} \to \mathcal{D}$ on the ensemble space such that, for every incoming ensemble ρ, there is a projective decomposition $\{P_a^\rho\}$ of the identity for which $\Psi(\rho) = \sum_a P_a^\rho \rho P_a^\rho$. The projective decomposition may depend on the incoming ensemble.*

We now prove the fact that connects the generator of a unitary group to a projective process.

Proposition 2 (The unitary resolvent is state-adaptive projective)**.** *Let K be a bounded self-adjoint operator on a Hilbert space $\mathcal{H}$ and let $\mathcal{L}_K$ be the generator on density operators given by $\mathcal{L}_K(\rho) = \frac{[K,\rho]}{\imath\hbar}$. For every $\tau > 0$, define*

$$\Psi_\tau = (I - \tau\mathcal{L}_K)^{-1}. \tag{3}$$

Then Ψ_τ is an affine completely positive trace-preserving map. Moreover, for every density operator ρ, there is a projective decomposition $\{P_a^\rho\}$ of the identity such that

$$\Psi_\tau(\rho) = \sum_a P_a^\rho \rho P_a^\rho. \tag{4}$$

Proof. Let $\mathcal{T}_t^K(\rho) = e^{tK/(\imath\hbar)}\rho e^{-tK/(\imath\hbar)}$. We first recall how the integral representation of the inverse arises in the scalar case. Since

$$\frac{d}{ds}\left(e^{-s}e^{as}\right) = -(1-a)e^{-s}e^{as}, \tag{5}$$

integration from zero to infinity gives

$$(1-a)\int_0^\infty e^{-s}e^{as}\,ds = -\left[e^{-s}e^{as}\right]_{s=0}^{s=\infty} = 1, \tag{6}$$

whenever the boundary term at infinity vanishes. Therefore $(1-a)^{-1} = \int_0^\infty e^{-s}e^{as}\,ds$.

We now replace the scalar a with the map $\tau\mathcal{L}_K$. Since $e^{\tau s\mathcal{L}_K}(\rho) = \mathcal{T}_{\tau s}^K(\rho)$, the corresponding candidate for the inverse is

$$\Psi_\tau(\rho) = \int_0^\infty e^{-s}\mathcal{T}_{\tau s}^K(\rho)\,ds. \tag{7}$$

To verify the identity directly, note that $\frac{d}{ds}\mathcal{T}_{\tau s}^K = \tau\mathcal{L}_K\mathcal{T}_{\tau s}^K$. Consequently,

$$\frac{d}{ds}\left(e^{-s}\mathcal{T}_{\tau s}^K\right) = -e^{-s}(I-\tau\mathcal{L}_K)\mathcal{T}_{\tau s}^K. \tag{8}$$

If σ denotes the integral in 7, we therefore have

$$\begin{aligned}(I-\tau\mathcal{L}_K)(\sigma) &= \int_0^\infty e^{-s}(I-\tau\mathcal{L}_K)\mathcal{T}_{\tau s}^K(\rho)\,ds \\ &= -\left[e^{-s}\mathcal{T}_{\tau s}^K(\rho)\right]_{s=0}^{s=\infty} \\ &= \rho.\end{aligned} \tag{9}$$

The boundary term at infinity vanishes because unitary evolution preserves the trace norm, while e^{-s} tends to zero; at $s = 0$, $\mathcal{T}_0^K = I$. This proves that the integral in 7 is $(I - \tau\mathcal{L}_K)^{-1}(\rho)$.

Equation 7 expresses Ψ_τ as a statistical mixture of unitary channels. Therefore it is affine, completely positive and trace preserving.

Set $\sigma = \Psi_\tau(\rho)$ and write its spectral decomposition as $\sigma = \sum_a \lambda_a P_a^\rho$, where equal eigenvalues are grouped into the same spectral projection. From the definition of the resolvent,

$$\rho = (I - \tau\mathcal{L}_K)(\sigma) = \sigma - \tau\frac{[K,\sigma]}{\imath\hbar}. \tag{10}$$

Since $\sigma P_a^\rho = P_a^\rho\sigma = \lambda_a P_a^\rho$, we have

$$P_a^\rho\frac{[K,\sigma]}{\imath\hbar}P_a^\rho = 0. \tag{11}$$

Consequently,

$$P_a^\rho\rho P_a^\rho = P_a^\rho\sigma P_a^\rho = \lambda_a P_a^\rho. \tag{12}$$

Summing over the spectral projections gives $\sum_a P_a^\rho\rho P_a^\rho = \sigma = \Psi_\tau(\rho)$. Thus the same map that averages the unitary trajectory acts on each incoming ensemble as a projective measurement adapted to that ensemble. □

The process is close to the corresponding unitary step. The following estimates will be useful. They are uniform over density operators and pure states.

Proposition 13 (Infinitesimal prediction)**.** *Under the assumptions of Proposition 2, let $M = \|K\|$. Then, as $\tau \to 0$,*

$$\|\Psi_\tau(\rho) - \mathcal{T}_\tau^K(\rho)\|_1 = O\left(\frac{\tau^2 M^2}{\hbar^2}\right). \tag{14}$$

If P is pure, let $F_\tau(P)$ be the spectral projector of $\Psi_\tau(P)$ associated with its largest eigenvalue $q_\tau(P)$. Then

$$1 - q_\tau(P) = O\left(\frac{\tau^2 M^2}{\hbar^2}\right), \qquad \|F_\tau(P) - \mathcal{T}_\tau^K(P)\|_1 = O\left(\frac{\tau^2 M^2}{\hbar^2}\right). \tag{15}$$

Proof. Both the resolvent and the unitary evolution have the same first-order expansion:

$$\Psi_\tau = I + \tau\mathcal{L}_K + O(\tau^2\|\mathcal{L}_K\|^2), \qquad \mathcal{T}_\tau^K = I + \tau\mathcal{L}_K + O(\tau^2\|\mathcal{L}_K\|^2). \tag{16}$$

Since $\|\mathcal{L}_K\| \le 2M/\hbar$ on trace-class operators, the first estimate follows. If P is pure, $\mathcal{T}_\tau^K(P)$ is also pure. The first estimate shows that $\Psi_\tau(P)$ differs from this rank-one projector only at second order. Standard perturbation of an isolated eigenvalue then gives the two remaining estimates: the largest eigenvalue differs from one at second order, and its spectral projector differs from $\mathcal{T}_\tau^K(P)$ at second order. Moreover, the proof of Proposition 2 gives $P_a^P P P_a^P = \lambda_a P_a^P$. Since the left-hand side has rank at most one, every P_a^P with $\lambda_a > 0$ has rank one. Thus $q_\tau(P)$ is precisely the probability of the corresponding pure measurement outcome. □

We will also use the following characterization of affine correspondences of quantum ensembles.

Proposition 17 (Affine pure-state correspondence)**.** *Let $\Phi : \mathcal{D}(\mathcal{H}) \to \mathcal{D}(\mathcal{H})$ be an affine map. If Φ maps the pure states bijectively onto the pure states, then there is either a unitary or an anti-unitary operator $W : \mathcal{H} \to \mathcal{H}$ such that $\Phi(\rho) = W\rho W^\dagger$ for every density operator ρ.*

Proof. Affinity extends Φ to a positive trace-preserving real-linear map on the self-adjoint trace-class operators. Since pure states are exactly the positive trace-one rank-one operators, the extension maps positive rank-one operators bijectively onto positive rank-one operators. The linear rank-one preserver theorem gives either $\Phi(\rho) = W\rho W^\dagger$ or, after choosing an orthonormal basis, $\Phi(\rho) = W\rho^\mathsf{T} W^\dagger$. Positivity and trace preservation make W an isometry, while surjectivity on pure states makes it surjective. The first case is therefore implemented by a unitary and the second by an anti-unitary. $\square$

3 Conditions

The following condition represents the standard structure of quantum states on Hilbert space.

Condition QST:H (Quantum states on Hilbert space)**.** *The state space of the system is represented by the projective space $\mathcal{P}(\mathcal{H})$ of a separable complex Hilbert space $\mathcal{H}$. The ensemble space is represented by the density operators $\mathcal{D}(\mathcal{H})$. Observables are represented by self-adjoint operators. The conditional probability is given by the Born rule $p(\psi|\phi) = \frac{\langle\phi|\psi\rangle\langle\psi|\phi\rangle}{\langle\psi|\psi\rangle\langle\phi|\phi\rangle}$. The entropy is given by the von Neumann entropy: $S(\rho) = -\operatorname{tr}(\rho\log\rho)$.*

We compare the usual mathematical characterization of unitary evolution with a physical characterization in terms of infinitesimal projective processes.

Condition DR-UNIT (Unitary evolution)**.** *Time evolution is represented by a strongly continuous one-parameter group of unitary operators. That is, $\Phi_t(\rho) = U_t\rho U_t^\dagger$, where $U_t : \mathcal{H} \to \mathcal{H}$, $U_0 = I$, $U_{t+s} = U_tU_s$ and $U_t^\dagger U_t = U_tU_t^\dagger = I$.*

Condition DR-PROJ (Projective evolution)**.** *Time evolution is the limit of repeated applications of an infinitesimal state-adaptive projective process that allows perfect prediction and perfect reconstruction. That is, evolution over elapsed time t is obtained by applying the same process of duration t/n, n times, in the limit $n \to \infty$. Moreover, the probabilities of the predicted and reconstructed state sequences tend to one.*

For the mathematical proof, the limit in DR-PROJ is understood in trace norm. A predicted state sequence is an outcome branch of the repeated projective processes, and its probability is the product of the conditional probabilities along that branch. Perfect prediction means that the branch converges to the evolved state while its probability tends to one; perfect reconstruction has the analogous meaning in the opposite time direction. As usual, time evolution over elapsed time composes over adjacent intervals and the infinitesimal limit is continuous.

4 Theorem

We now prove the equivalence.

Theorem 18. *Over QST:H, conditions DR-UNIT and DR-PROJ are equivalent.*

Proof. We first show that unitary evolution is a limit of state-adaptive projective processes.

DR-UNIT $\implies$ DR-PROJ. Let $U_t = e^{tH/(\imath\hbar)}$ be the strongly continuous one-parameter unitary group and let $\mathcal{T}_t(\rho) = U_t \rho U_t^\dagger$ be its action on density operators. Suppose first that H is bounded. Set $\mathcal{L}(\rho) = \frac{[H,\rho]}{\imath\hbar}$ and define the process of duration τ by

$$\Psi_\tau = (I - \tau\mathcal{L})^{-1}. \tag{19}$$

By Proposition 2, each Ψ_τ is state-adaptive projective. The resolvent exponential formula gives

$$\lim_{n\to\infty} \left(\Psi_{t/n}\right)^n (\rho) = \lim_{n\to\infty} \left(I - \frac{t}{n}\mathcal{L}\right)^{-n} (\rho) = e^{t\mathcal{L}}(\rho) = \mathcal{T}_t(\rho). \tag{20}$$

It remains to check perfect prediction. For a pure incoming state P, choose at each step the outcome $F_\tau(P)$ associated with the largest eigenvalue of $\Psi_\tau(P)$. Proposition 13 shows that, for some constant C independent of P and sufficiently small τ, the failure probability at one step is at most $C\tau^2\|H\|^2/\hbar^2$. With $\tau = t/n$, the probability that all n predictions succeed is therefore bounded below by

$$\left(1 - C\frac{t^2\|H\|^2}{n^2\hbar^2}\right)^n, \tag{21}$$

which tends to one. The selected state differs from the corresponding unitary step by the same second-order quantity. Since unitary conjugation is an isometry in trace norm, the errors of the n selected states accumulate to at most $O(t^2\|H\|^2/(n\hbar^2))$, which tends to zero. Thus the predicted state sequence converges to the unitary trajectory. Affinity is essential here: after one unconditioned step the output is mixed, but affinity guarantees that applying Ψ_τ to that mixture gives the mixture of the processes applied to its pure components. The outcome branches can therefore be followed consistently without choosing a particular decomposition of the mixed ensemble. Replacing H with $-H$ gives a state-adaptive projective process for the reconstructed sequence and proves perfect reconstruction.

Now let H be unbounded. The previous argument cannot be applied directly because its one-step estimates contain $\|H\|^2$. We therefore need bounded operators H_τ with two properties. First, their unitary evolutions must converge to the unitary evolution generated by H. Second, the one-step errors must still vanish after t/τ repetitions. Since the one-step errors are of order $\tau^2\|H_\tau\|^2$, their accumulated contribution is of order $t\tau\|H_\tau\|^2$. We must therefore arrange that H_τ approaches H while $\tau\|H_\tau\|^2$ tends to zero.

For each $\tau > 0$, choose a positive cutoff M_τ such that

$$M_\tau \longrightarrow \infty, \qquad \tau M_\tau^2 \longrightarrow 0. \tag{22}$$

Let $f_\tau(x) = \max\{-M_\tau, \min\{x, M_\tau\}\}$ and set $H_\tau = f_\tau(H)$ through the spectral calculus. Thus H_τ is bounded, $\|H_\tau\| \le M_\tau$ and $f_\tau(x) \to x$ for every real x. The first limit in 22 therefore ensures that $e^{tH_\tau/(\imath\hbar)}$ converges strongly to $e^{tH/(\imath\hbar)}$ for every fixed t. The second limit ensures that the accumulated errors vanish.

Define $\mathcal{L}_\tau(\rho) = \frac{[H_\tau,\rho]}{\imath\hbar}$ and

$$\Psi_\tau = (I - \tau\mathcal{L}_\tau)^{-1}. \tag{23}$$

Each Ψ_τ is again state-adaptive projective. For fixed t and $\tau = t/n$, Proposition 13 shows that the accumulated state error and the total failure probability are both $O(t\tau M_\tau^2/\hbar^2)$, which tends to zero by 22.

It remains to verify that the unconditioned process converges to the required evolution. The one-step difference between Ψ_τ and $e^{\tau\mathcal{L}_\tau}$ has the same second-order bound. Both maps contract trace distance, so a telescoping sum bounds the difference after n steps by n times the one-step difference. Consequently,

$$\left\| \left(\Psi_{t/n}\right)^n (\rho) - e^{t\mathcal{L}_{t/n}}(\rho) \right\|_1 \longrightarrow 0. \tag{24}$$

Strong convergence of $e^{tH_{t/n}/(\imath\hbar)}$ to $e^{tH/(\imath\hbar)}$ implies convergence of the corresponding conjugations on every trace-class operator, so

$$\left\| e^{t\mathcal{L}_{t/n}}(\rho) - \mathcal{T}_t(\rho) \right\|_1 \longrightarrow 0. \tag{25}$$

Combining the last two limits proves 20. Replacing H_τ with $-H_\tau$ gives the reconstructed sequence. Therefore the unitary evolution of an arbitrary self-adjoint generator satisfies DR-PROJ.

We now prove the converse.

DR-PROJ $\implies$ DR-UNIT. Write Φ_t for the limiting time evolution. Each finite process is affine, so every finite repetition is affine and its trace-norm limit Φ_t is affine.

Consider a pure initial state. At finite n, the predicted sequence identifies one branch of the repeated projective process. Affinity guarantees that the unconditioned repeated process is the mixture of all such outcome branches, independently of how any intermediate mixed ensemble is decomposed. The weight of the predicted branch is the product of its conditional probabilities. By perfect prediction, this weight tends to one and its final state tends to the predicted future state. The unconditioned output is therefore a mixture containing this branch with a weight tending to one, so its trace distance from the predicted pure state tends to zero. Consequently, Φ_t maps every pure initial state to a pure future state. Perfect reconstruction gives the converse correspondence: every pure future state reconstructs a unique pure initial state. Hence Φ_t maps the pure states bijectively onto the pure states.

Proposition 17 now shows that each Φ_t is implemented by either a unitary or an anti-unitary operator. Since evolution composes, $\Phi_t = \Phi_{t/2} \circ \Phi_{t/2}$. The composition is implemented by a unitary whether $\Phi_{t/2}$ is implemented by a unitary or an anti-unitary. Therefore every Φ_t is implemented by a unitary operator U_t.

Evolution over adjacent time intervals composes, while the use of the same duration-dependent process makes the limiting evolution depend only on elapsed time. Therefore $\Phi_0 = I$ and $\Phi_{t+s} = \Phi_t \circ \Phi_s$. Perfect reconstruction supplies the inverse evolution. The phases of the implementing unitaries may be chosen so that $U_0 = I$ and $U_{t+s} = U_t U_s$. Continuity of the infinitesimal limit gives strong continuity of U_t. Thus the evolution is represented by a strongly continuous one-parameter group of unitary operators, which is condition DR-UNIT. □

5 Additional remarks

The finite process is not unitary. The process Ψ_τ is generally irreversible and maps pure ensembles to mixed ensembles. Its inverse $I - \tau\mathcal{L}$ is not generally positive. Reconstruction uses the distinct process $(I + \tau\mathcal{L})^{-1}$, not the mathematical inverse of Ψ_τ. Only in the infinitesimal limit do prediction and reconstruction become mutually inverse unitary evolutions.

The relation to the unitary trajectory. Equation 7 shows that the finite process is not chosen arbitrarily: it is the ensemble obtained by evolving unitarily for an exponentially distributed duration with mean τ. The spectral projections of this time-averaged ensemble define the projective measurement that produces the same output directly from the incoming ensemble.

Why the process is state-adaptive. The map Ψ_τ is affine and independent of any decomposition of the incoming density operator. The projective decomposition that realizes it, however, is the spectral decomposition of $\Psi_\tau(\rho)$ and therefore depends on ρ. Different incoming ensembles generally undergo different projective measurements even though they are acted upon by the same affine process.

The qubit picture. Define the Cayley transform $\mathcal{C}_\tau = (I + \tau\mathcal{L})(I - \tau\mathcal{L})^{-1}$. A direct calculation gives $\Psi_\tau = (I + \mathcal{C}_\tau)/2$. Since $\mathcal{L}$ is skew with respect to the Hilbert–Schmidt inner product, $\mathcal{C}_\tau$ is orthogonal. On the Bloch ball it is a rotation, so Ψ_τ sends a Bloch vector to the midpoint between that vector and its rotated image. This is the geometric qubit construction in generator form.

Unbounded generators. Direct use of the resolvent of an unbounded generator still gives a completely positive trace-preserving map and the resolvent exponential formula. The bounded spectral truncations are used only to ensure that the total probability of the selected state sequence tends to one for every Hilbert-space state, including states outside the domain of the Hamiltonian.

6 Conclusion

Unitary evolution can be understood as a reversible quasi-static limit of projective processes. At every finite scale the process is a projective measurement adapted to the incoming ensemble and generally produces a mixture. As the duration tends to zero, one outcome becomes perfectly predictable, the corresponding state sequence converges to the unitary trajectory and the reverse sequence reconstructs the past with probability one. The deterministic and reversible unitary group is therefore recovered from an infinitesimal sequence of individually irreversible projective processes.